

Appendix 21

The independent boson theorem

In this appendix we provide a proof for the important formula (9.107). As a first step we prove that

$$\left\langle e^{\sum_i x_i \hat{b}_i^\dagger} e^{\sum_i y_i \hat{b}_i} \right\rangle = e^{\sum_i x_i y_i \langle \hat{b}_i^\dagger \hat{b}_i \rangle}, \quad (\text{A21.1})$$

where x_i and y_i are c-numbers and the thermal average is taken over a set of independent bosons with hamiltonian $\hat{H} = \sum_i \hbar \omega_i \hat{b}_i^\dagger \hat{b}_i$.

For simplicity, we consider here explicitly the case of a single boson mode. The generalization to many modes is trivial since different boson operators commute with each other. A possible way of carrying out the calculation of the above average takes advantage of the idea of *boson coherent states*. Accordingly, before proceeding any further we review the definition and some of the most relevant properties of these states. In the process we will also introduce some devices which will prove useful in performing some of the calculations in Chapter 10.¹

One defines the coherent boson state $|\phi\rangle$ as follows:

$$|\phi\rangle \equiv e^{\phi \hat{b}^\dagger} |0\rangle, \quad (\text{A21.2})$$

where ϕ is a complex number and $|0\rangle$ is the boson vacuum.

As it is readily verified, the coherent states are eigenstates of the destruction operator with eigenvalue ϕ , i.e. they satisfy the conjugate relations

$$\hat{b}|\phi\rangle = \phi|\phi\rangle, \quad (\text{A21.3})$$

and

$$\langle \phi | \hat{b}^\dagger = \langle \phi | \phi^*. \quad (\text{A21.4})$$

Evidently, they also satisfy the equations

$$\hat{b}^\dagger |\phi\rangle = \frac{\partial}{\partial \phi} |\phi\rangle, \quad (\text{A21.5})$$

and

$$\langle \phi | \hat{b} = \frac{\partial}{\partial \phi^*} \langle \phi |. \quad (\text{A21.6})$$

¹ For more details on the idea of coherent states the reader is referred to the book by Negele and Orland, 1988.

Since $\hat{b}$ is non-hermitian the following useful relation can be obtained using the identity (9.81):

$$e^{\alpha\hat{b}}|\phi\rangle = e^{\alpha\hat{b}}e^{\phi\hat{b}^\dagger}|0\rangle = e^{\alpha\phi}e^{\phi\hat{b}^\dagger}e^{\alpha\hat{b}}|0\rangle = e^{\alpha\phi}|\phi\rangle. \quad (\text{A21.7})$$

Clearly the $|\phi\rangle$'s are not mutually orthogonal and their overlap is immediately obtained from Eq. (A21.7):

$$\langle\phi|\phi'\rangle = e^{\phi^*\phi'}. \quad (\text{A21.8})$$

Even though non-orthogonal, the coherent states enjoy an important completeness relation that allows every state to be expressed as a linear combination of them. This completeness (or, in fact, overcompleteness) relation reads

$$\int_{-\infty}^{+\infty} \frac{d\phi_1 d\phi_2}{\pi} e^{-|\phi|^2} |\phi\rangle\langle\phi| = \hat{\mathbf{1}}, \quad (\text{A21.9})$$

where we have introduced the notation $\phi = \phi_1 + i\phi_2$, i.e. ϕ_1 and ϕ_2 are, respectively, the real and the imaginary parts of ϕ . Here $\hat{\mathbf{1}}$ denotes the identity operator in the Hilbert space.

Eq. (A21.9) can be proved by showing that its left-hand side commutes with both $\hat{b}$ and $\hat{b}^\dagger$ and therefore must be proportional to the identity operator. The proof of this is simple and relies on the fact that, according to Eqs. (A21.3)–(A21.6), the commutator of (say) $\hat{b}$ with $|\phi\rangle\langle\phi|$ is given by

$$[\hat{b}, |\phi\rangle\langle\phi|] \equiv \hat{b}|\phi\rangle\langle\phi| - |\phi\rangle\langle\phi|\hat{b} = \left(\phi - \frac{\partial}{\partial\phi^*}\right)|\phi\rangle\langle\phi|, \quad (\text{A21.10})$$

where we have also used the fact that ϕ and ϕ^* are independent and therefore the partial derivative with respect to ϕ^* can be commuted with $|\phi\rangle$. Then the integral on the left-hand side of Eq. (A21.9) is seen to commute with $\hat{b}$ since, upon integrating by parts with respect to both ϕ_1 and ϕ_2 , one is left with an expression proportional to $\left(\phi + \frac{\partial}{\partial\phi^*}\right)e^{-|\phi|^2}$ which in turn clearly vanishes. The same procedure can be used for the case of the creation operator. The correctness of Eq. (A21.9) can be finally verified by taking the expectation value of its left-hand side in the vacuum state $|0\rangle$, and showing that it equals 1. Thus, any state $|\psi\rangle$ can be written as a superposition of the states $|\phi\rangle$'s according to the equation

$$|\psi\rangle = \int_{-\infty}^{+\infty} \frac{d\phi_1 d\phi_2}{\pi} e^{-|\phi|^2} |\phi\rangle\langle\phi|\psi\rangle. \quad (\text{A21.11})$$

After this brief excursus into the properties of coherent states we are now ready for the proof of the independent boson theorem. The sought average is explicitly written as

$$\langle e^{\alpha\hat{b}^\dagger} e^{\beta\hat{b}} \rangle = \frac{\sum_{n=0}^{\infty} \langle n | e^{\alpha\hat{b}^\dagger} e^{\beta\hat{b}} e^{-\frac{\hbar\omega}{k_B T} \hat{b}^\dagger \hat{b}} | n \rangle}{\sum_{n=0}^{\infty} \langle n | e^{-\frac{\hbar\omega}{k_B T} \hat{b}^\dagger \hat{b}} | n \rangle}. \quad (\text{A21.12})$$

We begin by making use of Eq. (A21.11) to rewrite the numerator as

$$\sum_{n=0}^{\infty} \langle n | e^{\alpha\hat{b}^\dagger} e^{\beta\hat{b}} e^{-\frac{\hbar\omega}{k_B T} \hat{b}^\dagger \hat{b}} | n \rangle = \int \frac{d\phi_1 d\phi_2}{\pi} e^{-|\phi|^2} \sum_{n=0}^{\infty} \langle n | \phi \rangle \langle \phi | e^{\alpha\hat{b}^\dagger} e^{\beta\hat{b}} e^{-\frac{\hbar\omega}{k_B T} \hat{b}^\dagger \hat{b}} | n \rangle. \quad (\text{A21.13})$$

The sum over n can now be carried out exactly in view of the completeness relation $|\phi\rangle = \sum_{n=0}^{\infty} \langle n|\phi\rangle |n\rangle$. We are therefore left with a trace over coherent states:

$$\int \frac{d\phi_1 d\phi_2}{\pi} e^{-|\phi|^2} \langle \phi | e^{\alpha \hat{b}^\dagger} e^{\beta \hat{b}} e^{-\frac{\hbar\omega}{k_B T} \hat{b}^\dagger \hat{b}} | \phi \rangle. \quad (\text{A21.14})$$

Next we observe that, with the notation $z = e^{-\frac{\hbar\omega}{k_B T}}$, we can write

$$e^{-\frac{\hbar\omega}{k_B T} \hat{b}^\dagger \hat{b}} |\phi\rangle = e^{-\frac{\hbar\omega}{k_B T} \hat{b}^\dagger \hat{b}} e^{\phi \hat{b}^\dagger} |0\rangle = e^{-\frac{\hbar\omega}{k_B T} \hat{b}^\dagger \hat{b}} e^{\phi \hat{b}^\dagger} e^{\frac{\hbar\omega}{k_B T} \hat{b}^\dagger \hat{b}} |0\rangle = e^{z\phi \hat{b}^\dagger} |0\rangle = |z\phi\rangle, \quad (\text{A21.15})$$

where we have made use of the fact that $e^{\frac{\hbar\omega}{k_B T} \hat{b}^\dagger \hat{b}} |0\rangle = |0\rangle$, and have recognized the formal similarity between the operator product in the third expression from the left, and a “time evolution” with hamiltonian $\hat{H} = \hbar\omega \hat{b}^\dagger \hat{b}$ over an imaginary time interval $-i\frac{\hbar}{k_B T}$. Thus, finally, making use of Eqs. (A21.14), (A21.7), and Eq. (A21.8), the numerator of Eq. (A21.12) can be recast in the form

$$\int \frac{d\phi_1 d\phi_2}{\pi} e^{-|\phi|^2} e^{\beta\phi z} \langle \phi | \alpha + z\phi \rangle = \int \frac{d\phi_1 d\phi_2}{\pi} e^{-|\phi|^2(1-z)} e^{\alpha\phi^*} e^{\beta\phi z}. \quad (\text{A21.16})$$

Proceeding in exactly the same manner, the denominator of that equation can be rewritten as

$$\sum_{n=0}^{\infty} \langle n | e^{-\frac{\hbar\omega}{k_B T} \hat{b}^\dagger \hat{b}} | n \rangle = \int \frac{d\phi_1 d\phi_2}{\pi} e^{-|\phi|^2(1-z)}. \quad (\text{A21.17})$$

The integrals appearing in the last two equations are standard gaussian integrals, which can be done analytically. Accordingly, for the numerator one obtains the expression

$$(1-z)^{-1} e^{\frac{z\alpha\beta}{1-z}}, \quad (\text{A21.18})$$

while for the denominator one gets the well known expression for the partition function of non-interacting bosons, i.e.,

$$\frac{1}{1-z} = \frac{1}{1 - e^{-\frac{\hbar\omega}{k_B T}}}. \quad (\text{A21.19})$$

By taking the ratio of the last two equations, and noting that $\frac{z}{1-z} = n_B(\hbar\omega) = \langle \hat{b}^\dagger \hat{b} \rangle$, we arrive at Eq. (A21.1).

Finally, the disentangling lemma allows us to combine the two exponentials on the left-hand side of Eq. (A21.1) into a single one to get the following identity:

$$\left\langle e^{\sum_i (x_i \hat{b}_i^\dagger + y_i \hat{b}_i)} \right\rangle = e^{\sum_i x_i y_i \left\langle \hat{b}_i^\dagger \hat{b}_i - \frac{1}{2} [\hat{b}_i^\dagger, \hat{b}_i] \right\rangle}. \quad (\text{A21.20})$$

Since $x_i y_i \left\langle \hat{b}_i^\dagger \hat{b}_i - \frac{1}{2} [\hat{b}_i^\dagger, \hat{b}_i] \right\rangle = \frac{1}{2} \left\langle (x_i \hat{b}_i^\dagger + y_i \hat{b}_i)^2 \right\rangle$ (the average of $\hat{b}_i^2$ and $(\hat{b}_i^\dagger)^2$ being zero), we see that

$$\left\langle e^{\sum_i (x_i \hat{b}_i^\dagger + y_i \hat{b}_i)} \right\rangle = e^{\frac{1}{2} \sum_i \langle (x_i \hat{b}_i^\dagger + y_i \hat{b}_i)^2 \rangle}, \quad (\text{A21.21})$$

and Eq. (9.107) is proved.

It is easy to see that the theorem is valid even if the average is done in the thermal ensemble of the hamiltonian $\sum_i \hbar \omega_i \hat{\beta}_i^\dagger \hat{\beta}_i$, where the $\hat{\beta}_i$'s are related to $\hat{b}_i$ and $\hat{b}_i^\dagger$ by a linear unitary transformation. Indeed, expressing $\hat{b}$ and the $\hat{b}^\dagger$ in terms of $\hat{\beta}$ and $\hat{\beta}^\dagger$ amounts to a redefinition of the coefficients x_i and y_i on both sides of Eq. (A21.21), with no impact on the validity of that equation.